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is similar to iteration, but more powerful. Using recursion, we can solve a problem with a given input, by solving the instances of the problem with a part of the input.

Example 8.10. Customers are waiting in a line at a counter. The man at the counter wants to know how many customers are waiting in the line.

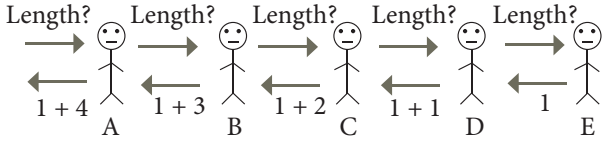

Figure 8.4: Length of a line

Instead of counting the length himself, he asks customer A for the length of the line with him at the head, customer A asks customer B for the length of the line with customer B at the head, and so on. When the query reaches the last customer in the line, E, since there is no one behind him, he replies 1 to D who asked him. D replies 1+1 = 2 to C, C replies 1+2 = 3 to B, B replies 1+3 = 4 to A, and A replies 1+4= 5 to the man in the counter

8.4.1 Recursive process

Example 8.10 illustrates a recursive process. Let us represent the sequence of 5 customers A, B, C, D and E as

[A,B,C,D,E]

The problem is to calculate the length of the sequence [A,B,C,D,E]. Let us name our solver length. If we pass a sequence as input, the solver length should output the length of the sequence.

length [A,B,C,D,E] = 5

Solver length breaks the sequence [A,B,C,D,E] into its first customer and the rest of the sequence.

first [A ,B,C,D,E] = A

rest [A ,B,C,D,E] = [B ,C,D,E]

To solve a problem recursively, solver length passes the reduced sequence [B,C,D,E] as input to a sub-solver, which is another instance of length. The solver assumes that the sub-solver outputs the length of [B,C,D,E], adds 1, and outputs it as the length of [A,B,C,D,E].

length [A,B,C,D,E] = 1 + length [B,C,D,E]

Each solver

1. receives an input,
2. passes an input of reduced size to a sub-solver,
3. receives the solution to the reduced input from the sub-solver, and produces the solution for the given input

as illustrated in Figure 8.5.

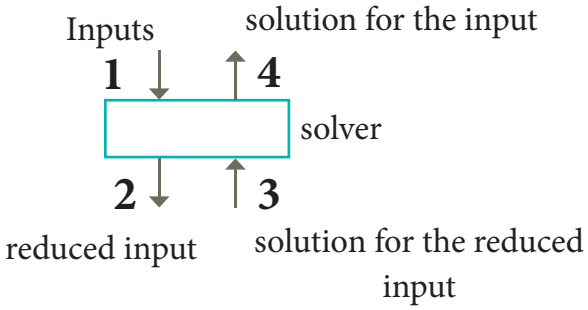

Figure 8.5: One instance of a solver in a recursive process

Figure 8.6 shows the input received and the solution produced by each